Assignment – 1.13

TCP/IP Suite Theory

Q - 1. Open Wireshark, capture network traffic while you refresh the Flipkart homepage, and identify one packet each belonging to the Application, Transport, Internet, and Network Access layers by examining their headers.

- **Wireshark Packet Analysis:**

  Aim: To capture network traffic while refreshing the Flipkart homepage and identify four network layers.

- **Observation Table:**

Layer	Protocol	Header information
Application	HTTP / TLS	Website data
Transport	TCP / UDP	Source and destination ports
Internet	IP	Source and destination IP addresses
Network Access	Ethernet	Source and destination MAC addresses

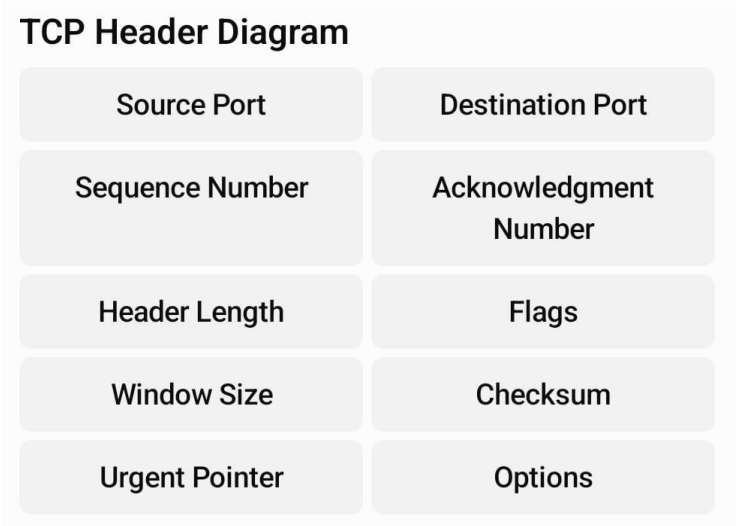
Q - 2. Create a table comparing the OSI and TCP/IP models, listing all layers side-by-side, and for each layer, give a real-world example from any app you use (e.g., WhatsApp message, Zomato order).

➢ Comparison of OSI and TCP/IP Models:

The **OSI model has 7 layers**, while the **TCP/IP model has 4 layers**. Both explain how data travels from one device to another over a network.

OSI Model (7 Layers)	TCP/IP Model (4 Layers)	Real-World Example
7. Application	Application	Sending a WhatsApp message
6. Presentation	Application	Encrypting WhatsApp messages
5. Session	Application	Maintaining a WhatsApp call
4. Transport	Transport	TCP/UDP carries data between devices
3. Network	Internet	IP address sends data to the destination
2. Data Link	Network Access	Wi-Fi transfers data using MAC addresses
1. Physical	Network Access	Wi-Fi radio signals carry data

<u>Q - 3.</u> Given a screenshot of a TCP segment header (provided by your trainer), label the fields that belong to the Transport layer and explain what each field does in 1-2 lines. Hint: Look for fields like Source Port, Destination Port, Sequence Number, etc.

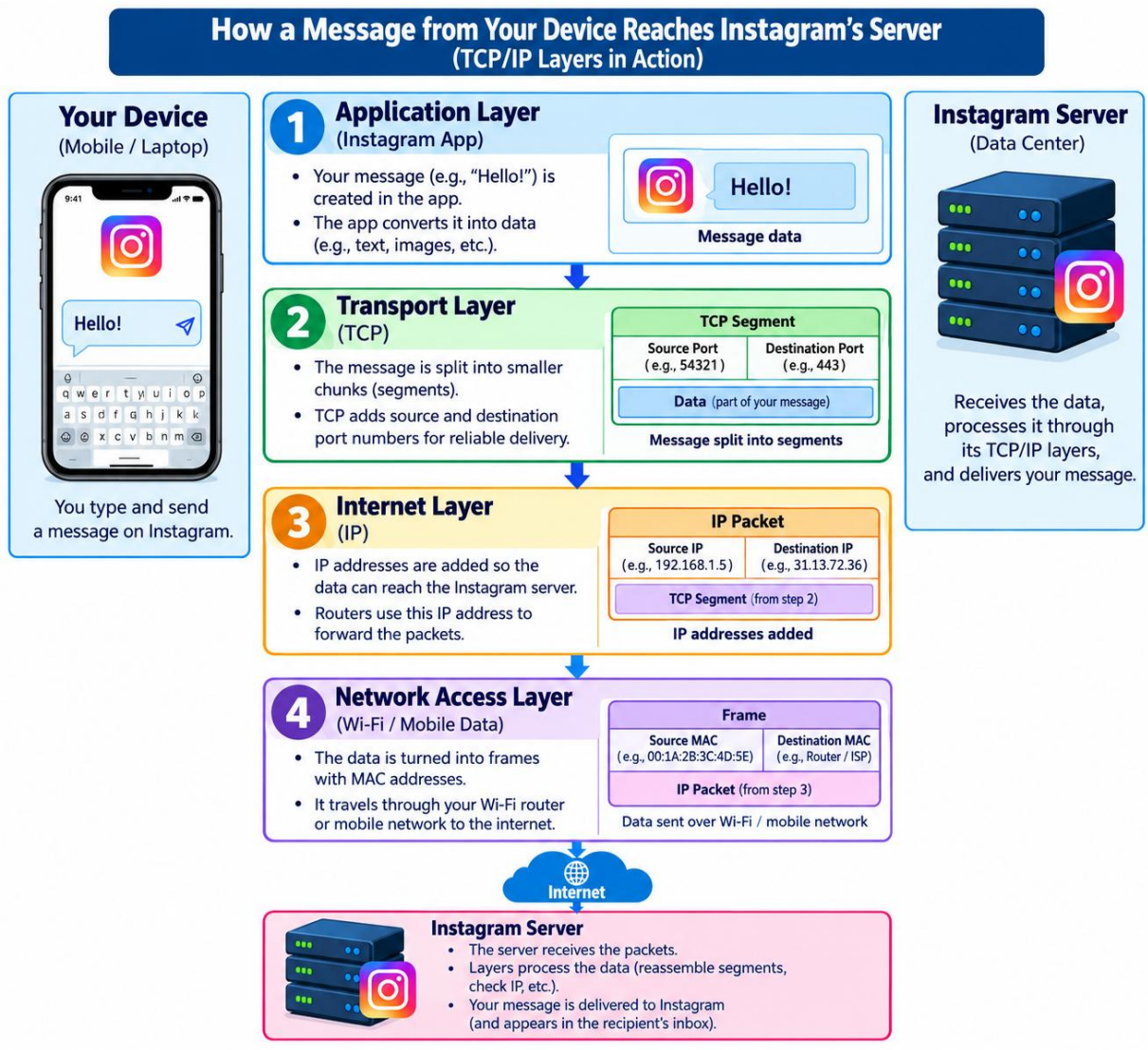


➢ <u>TCP Segment Header Fields:</u>

TCP Field	Function
1. **Source Port**	Identifies the sender's application.
2. **Destination Port**	Identifies the receiver's application.
3. **Sequence Number**	Keeps data in the correct order.

TCP Field	Function
4. **Acknowledgment Number**	Confirms received data.
5. **Header Length**	Shows the size of the TCP header.
6. **Flags**	Controls connection setup and closing.
7. **Window Size**	Controls how much data can be received.
8. **Checksum**	Checks for errors in data.
9. **Urgent Pointer**	Identifies urgent data.
10. **Options**	Contains extra TCP settings.

Q - 4. Draw a diagram showing how a message sent from your device to Instagram's server passes through each TCP/IP layer, and annotate what happens at each step using simple app-related terms (e.g., 'message split into packets', 'IP address added', etc.).



<u>Q - 5.</u> Pick any one packet you captured in Wireshark and write which TCP/IP layer it belongs to, how you identified it, and what real-world action in an app (like sending a WhatsApp message or booking a ticket on BookMyShow) it represents.

➢ **<u>Wireshark Packet Analysis:</u>**

- **Packet selected:** TCP packet
- **TCP/IP Layer:** Transport Layer
- **How I identified it:** I identified it by checking the TCP header in Wireshark, which contains the Source Port, Destination Port, and Sequence Number.
- **Real-world action:** When I send a WhatsApp message, TCP helps deliver the data reliably between my device and the server.